

MSC Timer App

Overview

The MSC Timer App is a centralized solution designed to track PNR-based activities using Office ID and Queue selection. It enables users to initiate and complete tasks through a structured start-end timer mechanism, automatically capturing duration and ensuring consistent, accurate logging through a SharePoint-backed system.

Business Context

The earlier Excel-based tracking system was successfully implemented and optimized to handle multi-user input, validation, and controlled data entry. It delivered reliable performance under normal conditions. However, due to Excel's file-based architecture, simultaneous write operations were limited, occasionally causing delays or requiring retries during concurrent usage scenarios.

PowerApps Solution – Key Benefits

1. Real-Time Multi-User Capability

- Multiple users can safely input and process data simultaneously
- Eliminates file-level locking issues from Excel
- Enables independent record-level operations
- Successfully validated under concurrent usage scenarios

2. Automated and Accurate Time Tracking

- Start Time is captured immediately at task initiation
- End Time and duration are calculated automatically
- Eliminates manual calculation errors
- Ensures consistent and accurate duration tracking

3. Advanced Validation & Business Rules

- PNR validation enforced (minimum 6 alphanumeric characters)
- Special handling for specific offices (auto "-" assignment)
- QC process ensures validation against existing PNR + Queue
- Duplicate detection with contextual user alerts

4. Structured Workflow with Session Control

- Start → End workflow strictly enforced
- Only one active session allowed per user at any time
- Prevents multiple device usage by the same user
- Ensures users complete one task before starting another
- Covers both loader and QC operations

5. Multi-Level QC Handling

- Supports up to **3 Quality Check levels (QC1, QC2, QC3)**
- QC entries dynamically attach to existing PNR + Queue records
- QC user is captured at Start and duration at End
- Prevents unique QC entries
- Enforces QC limit with validation controls

6. Centralized Data & Visibility

- All entries stored in SharePoint as a centralized data source
- Enables real-time tracking, reporting, and audit readiness
- Provides consistent and reliable data visibility across users

7. Strict Data Control

- Users are restricted to creating and completing entries only
- No direct editing or deletion of existing records
- Ensures data immutability and integrity
- Strengthens audit compliance and governance

Key Improvements over Excel

Area	Excel System	PowerApps Solution
Data Entry	Manual + controlled	Fully automated
Concurrency	Limited (file-based)	Verified real-time multi-user
Validation	Static rules	Dynamic runtime validation
Error Handling	Retry-based	Prevention-based
Session Control	Not available	Single active session enforced
QC Handling	Manual tracking	Structured multi-level QC
Data Integrity	Editable by users	Controlled & protected
Scalability	Limited	Scalable & structured

across:

- Multi-user concurrent usage (successfully passed)
- Single-session enforcement (multi-device prevention)
- QC multi-level handling (QC1 → QC3)
- Duplicate PNR scenarios
- Timer accuracy and duration calculations
- Data consistency and integrity in SharePoint

Conclusion

The MSC Timer App successfully transitions the process from a well-optimized Excel-based solution to a robust, scalable, and system-driven platform.

With successful **concurrency testing validation**, along with enhancements such as **session control and multi-level QC handling**, the application ensures reliable, conflict-free operations in real-world usage.

The solution is production-ready, offering improved accuracy, operational control, and strong data governance.

Testing & Validation Scope

The solution has been tested and validated

Power Apps | MSC_Timer_APP Subapp

CATHAY MSC_Queue_Log

Office ID: BOMCX0431

Queue: Q12C0

PNR: empty

Start Timer End Timer

Power Apps | MSC_Timer_APP Subapp

CATHAY MSC_Queue_Log

Office ID: BOMCX0431

Queue: Q12C0

PNR: empty

Start Timer End Timer

Microsoft Edge

Power Apps | MSC_Timer_APP Subapp

CATHAY MSC_Queue_Log

Office ID	Queue	PNR	Status	User Name	Action
BOMCX0431	Q12C0	empty	empty	BOMCX0431	

Power Apps | MSC_Timer_APP Subapp

CATHAY MSC_Queue_Log

Office ID: BOMCX0431

Queue: Q12C0

PNR: empty

Start Timer End Timer

Power Apps | MSC_Timer_APP Subapp

CATHAY MSC_Queue_Log

Office ID: QUALITY CHECK

Queue: Q12C0

PNR: empty

Start Timer End Timer

Power Apps | MSC_Timer_APP Subapp

CATHAY MSC_Queue_Log

Office ID: QUALITY CHECK

Queue: SECURITY REPORT QC

PNR: empty

Start Timer End Timer

CATHAY

MSC Queue Log

Office_ID

incoming

Queue

SH

PNR

Start Timer

End Timer

CATHAY

MSC Queue Log

Office_ID

Q12C0

Queue

Q12C0

PNR

mnoop2

Start Timer

End Timer

My lists

MSC_Queue_Log

All Items

	Office_ID	Queue	PNR	Start_Time	End_Time	Duration	User_Name	QC1_user	QC1_duration	QC2_user	QC2_duration	QC3_user	QC3_duration
	BOMCX0431	Q12C0	QWERTY	6/19/2026 8:37 PM	6/19/2026 8:37 PM	00:00:16	Rahul Goraksha	Rahul Goraksha	00:00:05	Rahul Goraksha	00:00:04	Rahul Goraksha	00:00:05
	BOMCX0431	Q12C0		6/19/2026 9:15 PM	6/19/2026 9:15 PM	00:00:04	Rahul Goraksha						
	BOMCX0431	Q30C1	QWERTY	6/19/2026 9:28 PM	6/19/2026 9:28 PM	00:00:04	Rahul Goraksha	Rahul Goraksha	00:00:05	Rahul Goraksha	00:13:46	Rahul Goraksha	00:00:11
	BOMCX0431	Q12C0	LOKE123	6/22/2026 4:46 PM	6/22/2026 4:46 PM	00:00:02	Lokesh Poojary						
	BOMCX0431	Q12C0	MNOP14	6/22/2026 4:42 PM	6/22/2026 4:42 PM	00:00:01	Rahul Goraksha	Rahul Goraksha	00:00:03	Rahul Goraksha	00:00:03	Rahul Goraksha	00:00:02
	REPORTS	RP		6/20/2026 12:32 AM	6/20/2026 12:32 AM	00:00:01	Rahul Goraksha						
	REPORTS	RP		6/19/2026 9:24 PM	6/19/2026 9:24 PM	00:00:03	Rahul Goraksha						
	REPORTS	RP	TIMER REPORT_1	6/19/2026 9:32 PM	6/19/2026 9:33 PM	00:00:42	Rahul Goraksha						
	REPORTS	RP	MSC REPORT	6/19/2026 9:31 PM	6/19/2026 9:31 PM	00:00:01	Rahul Goraksha						
	BOMCX0431	Q12C0	CHAP4321	6/22/2026 4:46 PM	6/22/2026 4:46 PM	00:00:04	Chandrasekhar Padashetty						
	TRAINING	HUB		6/19/2026 9:15 PM	6/19/2026 9:15 PM	00:00:01	Rahul Goraksha						
	TRAINING	OPERATIONAL		6/19/2026 9:30 PM	6/19/2026 9:30 PM	00:00:01	Rahul Goraksha						
	LF	MONTHLY REPORT	THIS IS A NICE MAIL	6/19/2026 9:22 PM	6/19/2026 9:22 PM	00:00:03	Rahul Goraksha						
	TRAINING	OPERATIONAL		6/19/2026 11:57 PM	6/19/2026 11:57 PM	00:00:03	Rahul Goraksha						
	SHIFT HANDLING	SH		6/19/2026 9:30 PM	6/19/2026 9:30 PM	00:00:01	Rahul Goraksha						

CATHAY

MSC_Queue_Log

Office_ID

QUALITY CHECK

Queue

Q12C0

PNR

mnoop2

Start Timer

End Timer